

**Dissertation:
Financial Risk Management**

Course No. : AIMLCZG628T

Course Title: Dissertation

Dissertation Work Done by:

Student Name: Dhairya Mihir Shah

BITS ID: 2024AB05218

Degree Program: M.Tech. AI/ML

**Research Area: Explainable Artificial Intelligence
for Financial Risk Management**

Dissertation / Project Work carried out at:

(Infosys Ltd - Pune, India -411057)



**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE,
PILANI
VIDYA VIHAR, PILANI, RAJASTHAN -**

333031.

(June 2026)

(June 2026)	1
1. Abstract	3
2. Dataset Selection and Analysis.....	3
3. Initial Research Direction	4
4. Research Pivot and Objective Refinement.....	4
5. Feature Store Architecture.....	4
6. Architecture Summary and Workflow Explanation	5
.....	8
7. Origination Feature Selection	9
8. Dataset Engineering Decisions	10
9. Modeling Dataset Strategy.....	11
10. Loan Amount Recommendation Model.....	11
11. Baseline Model: Linear Regression	12
12. Advanced Model: CatBoost.....	12
13. Feature Importance Findings	13
14. User Interface Development	13
15. Conclusion	14
16. Plan of Work.....	14
17. LiteratureReferences.....	15
Particulars of the Supervisor and Examiner	16
Information about the Supervisor:	17
AIMLCZG628T: Financial Risk Management	17

1. Abstract

The original objective of the dissertation was to build an Explainable Financial Risk Management platform capable of supporting lending decisions through transparent machine learning models. The initial scope included credit risk prediction, serviceability assessment, loan recommendation and explainability through SHAP and LIME.

During the implementation phase, it became evident that the Freddie Mac dataset contains two distinct categories of information:

- Origination Data
- Performance Data

Origination data represents information available when a loan is sanctioned, while performance data contains borrower behaviour after the loan has been issued.

This distinction significantly influenced the research direction and resulted in refinement of the project objectives.

2. Dataset Selection and Analysis

The Freddie Mac Single Family Loan Level Dataset was selected because it provides both origination and performance information at a loan level.

The dataset used in this research consists of all four quarters of 2018:

- 2018Q1 : 296,816 loans
- 2018Q2 : 364,502 loans
- 2018Q3 : 336,669 loans
- 2018Q4 : 287,447 loans
- Total Records : 1,285,434

The scale of this dataset is substantially larger than datasets commonly used in academic projects and therefore provides a realistic representation of industrial lending environments.

The dataset was initially loaded from quarterly origination files and subsequently merged into a unified master dataset.

3. Initial Research Direction

The first implementation focused on Serviceability Assessment.

A serviceability score was engineered using origination attributes and performance indicators. The intention was to estimate a borrower's repayment capability.

However, after detailed analysis of the Freddie Mac data structure, a conceptual issue was identified.

The serviceability score was being calculated using variables derived from future loan performance.

Examples included:

- Delinquency behaviour
- Loan ageing
- Remaining term evolution
- Interest rate movement

These variables are only available after loan origination and therefore cannot be used for lending recommendations at application time.

This observation led to a reassessment of the research objectives.

4. Research Pivot and Objective Refinement

A major design decision was taken to separate the research into two independent tracks.

- Track A: Origination-Based Recommendation Models
- Track B: Performance-Based Risk Models

The rationale behind this separation is that lending recommendations must only use information available at origination, whereas borrower stress and delinquency prediction can leverage future performance information.

This separation resulted in a cleaner and more realistic research design.

5. Feature Store Architecture

A version-controlled feature store was designed and implemented.

The purpose of the feature store is to provide reproducible datasets for different research objectives while maintaining complete traceability.

- Feature Store Version 1 focused on serviceability and stress prediction.
- Feature Store Version 2 introduced diagnostic improvements and dataset partitioning.
- Feature Store Version 3 was redesigned specifically for lending recommendation objectives.

Separate feature stores were created for:

- Loan Amount Recommendation
- Interest Rate Recommendation
- Loan Term Recommendation

This design allows independent experimentation without affecting other research objectives.

6. Architecture Summary and Workflow Explanation

The proposed Explainable Financial Risk Management Platform consists of two tightly integrated processing environments: an Offline Training Pipeline and an Online Recommendation Pipeline. This separation ensures that model development, validation and governance activities remain independent from real-time prediction services while maintaining consistency between training and inference.

- Offline Training Pipeline

The offline training pipeline is responsible for transforming raw Freddie Mac mortgage data into production-ready machine learning models. The process begins with the ingestion of quarterly Freddie Mac origination datasets covering all four quarters of 2018. These datasets collectively contain approximately 1.28 million mortgage records. Following

ingestion, extensive validation procedures are performed to identify missing values, data quality issues, incorrect data types and outlier observations.

Once validation is complete, feature engineering is performed. This stage generates machine learning features from origination variables such as credit score, debt-to-income ratio, loan-to-value ratio, combined loan-to-value ratio, property characteristics, and geographical information. The engineered features are stored in a version-controlled feature store to ensure reproducibility and traceability of experiments.

The feature store is subsequently used to create modeling datasets for different research objectives. Separate datasets are maintained for loan amount prediction, interest rate recommendation, loan term prediction, and future serviceability analysis.

The modeling datasets are partitioned into training, validation, and testing datasets using an 80-10-10 strategy. Machine learning algorithms including Linear Regression and CatBoost are then trained and evaluated using these datasets.

Following model training, explainability artifacts are generated. These include global feature importance rankings, SHAP explanations, and model performance reports. Trained models together with preprocessing metadata are stored within a model registry, allowing future deployment and governance activities.

- Online Recommendation Pipeline

The online recommendation pipeline represents the real-time decision support system that will be used by business users.

The process begins when a user enters borrower and property information through the recommendation portal. Inputs include borrower characteristics such as credit score, debt-to-income ratio and number of borrowers, along with property information such as property type, occupancy status, state, MSA and loan characteristics.

Before prediction occurs, the input data passes through an input validation layer. This layer performs range validation, mandatory field checks and datatype verification. For example, credit scores must fall within valid ranges, while LTV and CLTV values must satisfy business constraints.

After validation, the data undergoes cleaning and standardization. Missing values are

handled, categorical values are standardized and numerical fields are converted into the same format used during model training.

The cleaned data is then processed through an online feature engineering layer. This stage replicates the transformations applied during offline training to ensure that incoming records are represented identically to the data used during model development.

A feature alignment step follows. This component guarantees that feature ordering, data types and column structures exactly match those expected by the trained models. This prevents training-serving skew and ensures consistency between offline and online environments.

The aligned feature vector is subsequently passed to multiple prediction models. These models generate recommendations for loan amount, interest rate and loan term. Additional models may be integrated in future iterations of the platform.

Once predictions are generated, the explainability layer calculates local SHAP explanations for the specific recommendation. These explanations identify the most influential factors contributing to the model output and provide transparency for business users.

The final recommendation package is presented through the user interface and includes recommended loan amount, recommended interest rate, recommended loan term and a ranked explanation of the key drivers behind the recommendation.

- Pipeline Monitoring Dashboard

A separate monitoring dashboard has been designed to support model governance and operational oversight.

The dashboard provides visibility into:

- Dataset statistics
- Pipeline execution status
- Data quality reports
- Model performance metrics

- Feature importance rankings
- SHAP explainability reports
- Recommendation activity logs

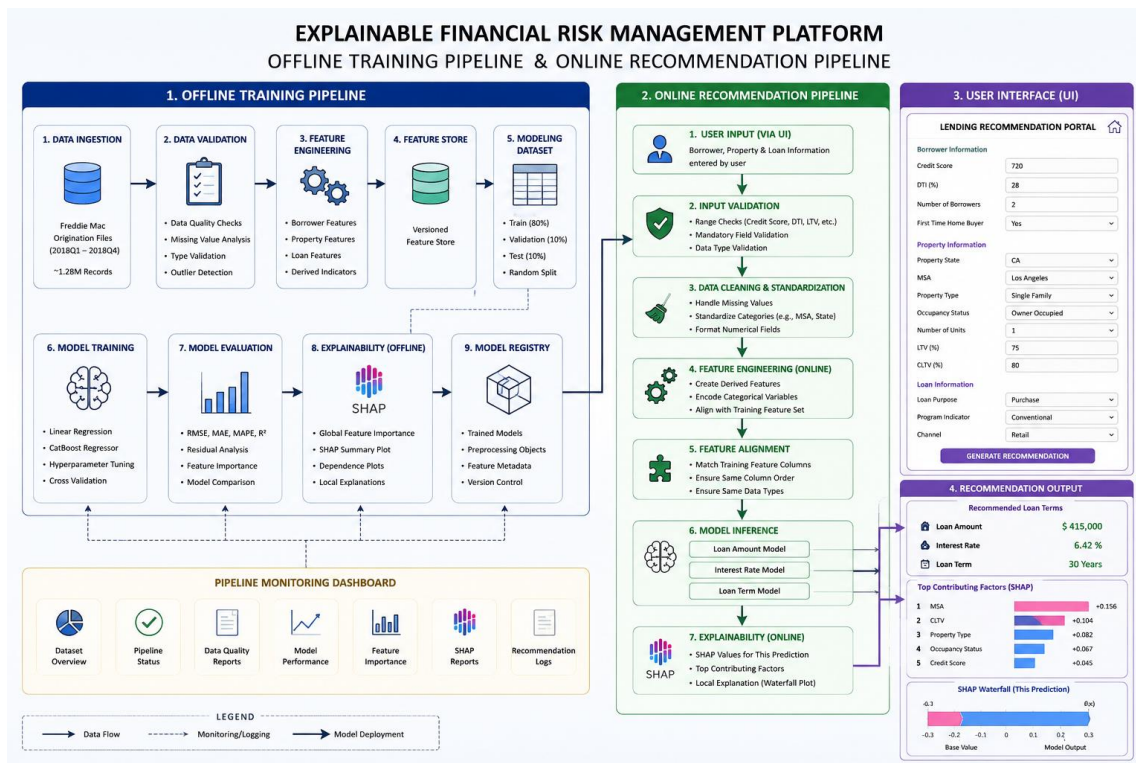
This dashboard serves as a centralized control panel for monitoring the health, performance and transparency of the overall financial risk management platform.

- Benefits of the Architecture

The architecture provides several advantages:

- Separation of training and inference workloads.
- Consistent feature engineering across offline and online environments.
- Version-controlled feature store for reproducibility.
- Explainability integrated directly into the prediction workflow.
- Governance-ready monitoring and reporting capabilities.
- Scalability for future recommendation and risk assessment models.

The resulting system represents a complete end-to-end framework that combines data engineering, machine learning, explainable AI and user-facing decision support capabilities within a single integrated platform.



7. Origination Feature Selection

A significant amount of effort was spent analyzing which origination variables should be retained.

Several iterations of feature selection were performed.

The final origination feature set contains:

- credit_score
- first_time_homebuyer_indicator
- dti
- num_borrowers
- occupancy_status
- property_type
- property_state
- msa
- num_units

- ltv
- cltv
- mortgage_insurance_pct
- channel
- loan_purpose
- program_indicator
- property_valuation_method
- interest_only_indicator
- mortgage_insurance_cancellation_indicator
- quarter
- quarter_num

Variables such as normalized versions of the same attributes were intentionally removed because they duplicated information already contained within the original variables.

Year was removed because all observations belonged to 2018 and therefore contributed no predictive information.

8. Dataset Engineering Decisions

Several design decisions were taken during feature preparation.

- Missing Values

MSA contained approximately 104,000 missing observations.

Rather than removing these records, an UNKNOWN category was introduced because geographic information was expected to be highly predictive.

- Numerical Variables

Credit score, DTI, CLTV and LTV contained a small number of missing observations.

Median imputation was selected to preserve dataset size while minimising distortion.

- Categorical Variables

Categorical variables were converted to string format to support CatBoost and One Hot Encoding workflows.

This step became necessary after training failures caused by mixed numeric and string

data types.

9. Modeling Dataset Strategy

The initial design used temporal splitting:

Training:

2018Q1 + 2018Q2

Validation:

2018Q3

Testing:

2018Q4

This approach was suitable for forecasting style problems.

However, after the research pivot, the objective changed from future prediction to recommendation learning.

Since the goal became learning lender decision patterns rather than forecasting future borrower behavior, a random split strategy was adopted.

Final Split:

Training: 80%

Validation: 10%

Testing: 10%

The resulting datasets contained:

Training: 1,028,347 observations

Validation: 128,543 observations

Testing: 128,544 observations

Quarter distributions were verified after splitting and remained consistent with the original population.

10. Loan Amount Recommendation Model

The first completed recommendation model predicts approved loan amount.

- Target Variable:

original_upb

- Transformation:

$\log(1 + \text{original_upb})$

The logarithmic transformation was introduced because the original loan amount distribution exhibited substantial skewness.

This transformation produced a more stable regression target.

11. Baseline Model: Linear Regression

The first benchmark model used Linear Regression.

Results:

RMSE:0.3926

MAE:0.3060

MAPE: 2.53%

R²: 0.5031

This result demonstrated that approximately half of the variation in approved loan amounts could be explained using borrower, property and underwriting attributes.

12. Advanced Model: CatBoost

A CatBoost Regressor was subsequently implemented.

Configuration:

Iterations:1000

Depth:8

Learning Rate:0.03

Results:

RMSE:0.3702

MAE: 0.2882

MAPE:2.38%

R^2 : 0.5582

The CatBoost model improved performance across all evaluation metrics.

The improvement confirms the presence of non-linear relationships within lender decision patterns.

13. Feature Importance Findings

One of the most significant findings emerged from CatBoost feature importance analysis.

Top Variables:

MSA: 31.86%

CLTV: 15.13%

Property Type: 12.09%

Property State: 8.18%

Occupancy Status: 7.10%

Number of Borrowers: 5.44%

LTV: 5.34%

Credit Score: 2.57%

An unexpected observation was the relatively low importance of credit score compared to geographical and collateral-related variables.

This suggests that approved loan amount is more strongly influenced by location and property characteristics than borrower credit quality.

14. User Interface Development

Unlike many academic projects that stop at model development, this dissertation includes deployment-oriented components.

Two user interfaces are currently being developed.

Pipeline Dashboard

Displays:

- Dataset Statistics
- Pipeline Status

- Training Metrics
- Diagnostics Results
- Explainability Outputs

Lending Recommendation Portal

Accepts borrower and property characteristics as input and generates:

- Recommended Loan Amount
- Recommended Interest Rate
- Recommended Loan Term
- Explainability Summary

Screenshots of both interfaces will be included in the final submission.

15. Conclusion

The dissertation has evolved significantly since its initial conception. Through detailed exploration of the Freddie Mac dataset, a clear distinction was established between origination information and performance information, leading to a refined research design. A scalable data pipeline, version-controlled feature store, diagnostics framework and recommendation modeling environment have been successfully implemented. Preliminary results demonstrate that lender decisions can be learned from origination characteristics and that geographic and collateral variables play a larger role in loan amount determination than initially expected. The remaining work focuses on explainability, recommendation refinement and user interface integration.

16. Plan of Work

Phases	Start Date-End Date	Work to be done	Status
Dissertation Outline	—05 th May 2026 10 th May 2026	Literature Review and prepare Dissertation Outline	Completed

Data Gathering, Research & Design	_11 th May 2026	Data Gathering, Research and Design Activity.	Completed
Development	_20 th May 2026 - 15 th June 2026	Development Activity including UI screen.	Completed
Testing	_20 th June 2026 - 01 th July 2026	Testing of the Models used for Credit Decisioning and Scoring.	In Progress
Research Explainability and Documentation	_02 th July 2026 - 20 th July 2026	Document Explainable AI Features and Evaluation of Model Bias.	In Progress
Dissertation Review	_21 st July 2026 - 28 th July 2026	Submit Dissertation to & Supervisor Examiner feedback	
Submission	28 th July 2026 - 02 nd August 2026	Final Review and submission of Dissertation	

17. Literature References

1. Bussmann, N., Giudici, P., Marinelli, D., & Papenbrock, J. (2021). Explainable Machine Learning in Credit Risk Management. *Computational Economics*, 57(1), 203–216.
2. Misheva, B., Osterrieder, J., Hirs, A., Kulkarni, O., & Lin, S. F. (2021). Explainable AI in Credit Risk Management. *arXiv:2103.00949*.
3. Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems (NeurIPS)*.
4. Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased Boosting with Categorical Features. *Advances in Neural Information Processing Systems (NeurIPS)*.

5. Dorogush, A. V., Ershov, V., & Gulin, A. (2018). CatBoost: Gradient Boosting with Categorical Features Support. arXiv:1810.11363.
6. Wang, H., & Cheng, L. (2021). CatBoost Model with Synthetic Features in Application to Loan Risk Assessment of Small Businesses. arXiv:2106.07954.
7. Freddie Mac. Single-Family Loan-Level Dataset. Federal Home Loan Mortgage Corporation.
8. Freddie Mac. Single-Family Loan-Level Dataset User Guide and Data Dictionary. Federal Home Loan Mortgage Corporation.
9. Ozturkkal, B., & Wahlström, R. R. (2025). Explaining Mortgage Defaults Using SHAP and LASSO. Computational Economics.
10. European Banking Authority (EBA). (2021). Report on the Use of Machine Learning for Internal Ratings Based Models.
11. OECD. (2024). OECD Framework for Artificial Intelligence Governance.
12. Basel Committee on Banking Supervision. (2023). Principles for the Effective Management and Supervision of Financial Risks.

Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Deval Dhyani	Shailesh Kumar
Qualification	Bachelors of Engineering	Bachelors of Engineering
Designation	Lead Consultant	Lead Consultant
Employing Organization and Location	Infosys Ltd, Sydney, Australia	Infosys Ltd, Bangalore, India

Phone No.(with STD Code)	+61 421417376	+91 9108958150
Email Address	Deval_dhyani@infosys.com	Shailesh_kumar06@infosys.com

Information about the Supervisor:

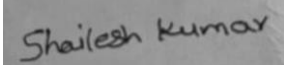
BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2021-2022
AIMLCZG628T: Financial Risk Management

STUDENT ID No.	2024AB05218
NAME OF THE STUDENT	Dhairya Mihir Shah
STUDENT'S EMAIL ADDRESS	2024AB05218@wilp.bits-pilani.ac.in
STUDENT'S EMPLOYING ORGANIZATION & LOCATION	Infosys Ltd, Pune
SUPERVISOR'S NAME	Deval Dhyani
SUPERVISOR'S EMPLOYING ORGANIZATION & LOCATION	Infosys Ltd, Sydney, Australia
SUPERVISOR'S EMAIL ADDRESS	Deval_dhyani@infosys.com
ADDITIONAL EXAMINER'S NAME	Shailesh Kumar
ADDITIONAL EXAMINER'S EMPLOYING ORGANIZATION & LOCATION	Infosys Ltd, Bangalore, India
ADDITIONAL EXAMINER'S EMAIL ADDRESS	Shailesh_kumar06@infosys.com
DISSERTATION / PROJECT / PROJECT WORK TITLE	Financial Risk Management

Please prepare the outline as a separate document with the following sections along with the above

identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing organization)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

		
Signature of Student	 Signature of Supervisor	Signature of Additional Examiner
Name: Dhairya Mihir Shah	Name: Deval Dhyani	Name: Shailesh Kumar

